

FH9261-G3JH

Single-Cell Lithium-Ion/Lithium-Polymer Battery

Protection IC

Overview

The FH9261-G3JH incorporates a high-precision voltage detection circuit and a delay circuit. By monitoring the battery's voltage and current, it provides protection

against overcharge

, over-discharge, and overcurrent protection by monitoring the battery's voltage and current. It is suitable for protection circuits of single-cell lithium-ion/lithium-polymer rechargeable batteries.

Features

- 1) High-precision voltage detection
 - Overcharge protection voltage 4.280 V
 - Overcharge release voltage 4.080 V
 - Over-discharge protection voltage 3.000 V
 - Over-discharge release voltage 3.000 V
 - Discharge overcurrent protection voltage 0.080 V
 - Short-circuit protection voltage 1.000 V
 - Charge overcurrent protection voltage -0.100 V
- 2) Charger detection and load detection functions
- 3) 0V battery charging function Allowed
- 4) Sleep function Available
- 5) Conditions for Clearing Overcurrent Discharge Disconnect the load
- 6) Voltage for clearing discharge overcurrent condition V_{DIOV}
- 7) Low current consumption
 - During operation 1.5 μA (typical) ($T_a = +25^\circ C$)
 - In sleep mode 0.1 μA (Max) ($T_a = +25^\circ C$)
- 8) Lead-free, halogen-free

Applications

- Single-cell lithium-ion/lithium-polymer rechargeable batteries

Package Pinout and Silkscreen Information

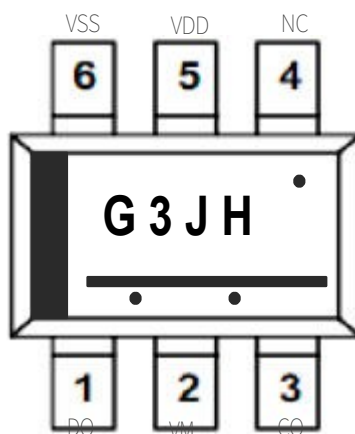


Figure 1

Note: The dots above and below the product model number are for internal identification purposes; their positions are subject to change.

Pin Description

Protection IC

SOT23-6 Package

Pin	Symbol	Description
1	DO	Terminal for connecting the MOSFET gate for discharge control
2	VM	Overcurrent detection input terminal, charger detection terminal
3	CO	MOSFET gate connection terminal for charge control
4	NC	No connection
5	VDD	Power supply, positive power input terminal
6	VSS	Ground terminal, negative power supply input terminal

Table 1

System Functional Block Diagram

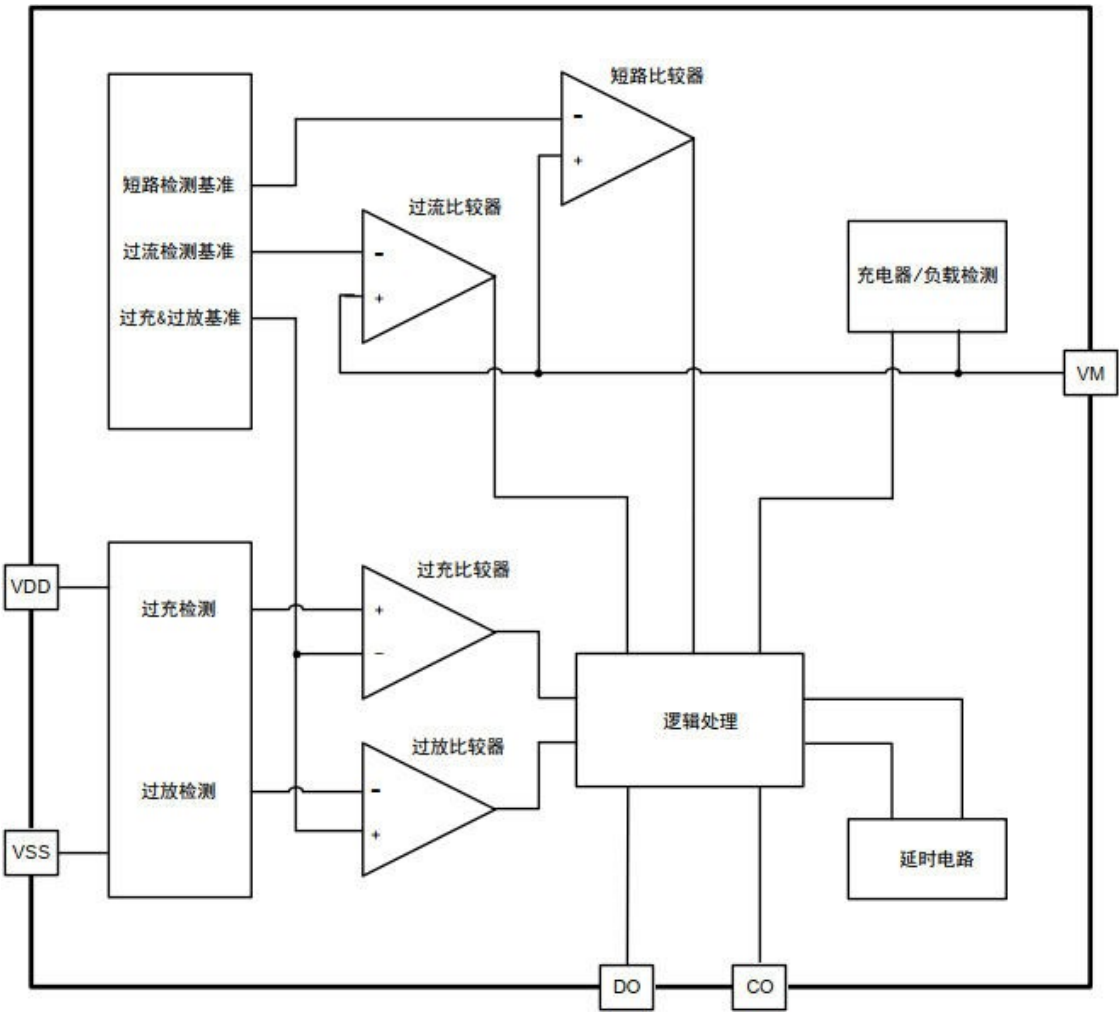


Figure 2

Product Model

Protection IC

Product Name	Overcharge protection voltage VOC	Overcharge release voltage VOCR	Over-discharge protection voltage VOD	Over-discharge release voltage VODR	Over-discharge protection voltage VEC	Short-circuit protection voltage VSHORT	Charge overcurrent protection voltage VCHA	Overcharge lockout	Over-discharge lockout
FH9261-G3JH	4.280 V	4.080 V	3.000 V	3.000 V	0.080 V	1.000 V	-0.100 V	Y	Y

Table 2

Absolute Maximum Ratings

(Unless otherwise specified: Ta = +25°C)

Item	Symbol	Absolute Maximum Ratings	Unit
Input voltage between VDD and VSS	VDD	VSS-0.3 to VSS+8.0	V
VM Input terminal voltage	VM	VDD-12 ~ VDD+0.3	V
CO output terminal voltage	VCO	VM-0.3 ~ VDD+0.3	V
DO output terminal voltage	VDO	VSS-0.3 ~ VDD+0.3	V
Operating temperature range	TOFR	-40 to +85	°C
Storage Temperature Range	TSTG	-55 to +125	°C

Table 3

Note: Applying a voltage exceeding the absolute maximum rating may cause irreversible damage to the chip.

Electrical Characteristics

Protection IC

(Unless otherwise specified: Ta = +25°C)

Item		Symbol	Test Conditions	Minimum	Typical Value	Maximum	Unit
Chip Supply Voltage		VDD	-	1.0	-	7.5	V
Normal operating current		IVDD	VDD=3.5V	0.9	1.5	3.0	μA
Sleep current consumption		IOPED	VDD = 1.5 V	-	-	0.1	μA
Overcharge	Protection voltage	VOC	VDD = 3.5~4.8 V	4.255	4.280	4.305	V
	Release voltage	VOCR	VDD = 4.8 → 3.5 V	4.030	4.080	4.130	V
	Protection delay	TOC	VDD = 3.5~4.8 V	500	1000	2000	ms
Over-discharge	Protection voltage	VOD	VC5=3.5→2.0V	2.920	3.000	3.080	V
	Release voltage	VODR	VDD = 2.0~3.5 V	2.900	3.000	3.100	V
	Protection delay	TOD	VDD = 3.5 → 2.0 V	64	128	256	ms
Discharge overcurrent	Protection voltage	VEC	VM-VSS=0→0.20V	0.065	0.080	0.095	V
	Protection delay	TEC	VM-VSS=0→0.20V	4	8	16	ms
Charging overcurrent	Protection voltage	VCHA	VSS-VM=0→0.30V	-0.130	-0.100	-0.070	V
	Protection delay	TCHA	VSS-VM=0→0.30V	4	8	16	ms
Short-circuit	Protection voltage	VSHORT	VM - VSS = 0 to 1.5 V	0.700	1.000	1.300	V
	Protection delay	TSHORT	VM -VSS=0→1.5V	125	250	500	μs
0V Charging Charger Start Voltage		V0VCH	0V Battery Charging Enabled	0.0	0.7	1.5	V

Table 4

■ Function Description

Protection IC

1. Normal Operating Condition

The IC continuously monitors the battery voltage between the VDD and VSS terminals, as well as the voltage between the VM and VSS terminals, to control charging and discharging. When the battery voltage is above the over-discharge protection voltage (V_{OD}) and below the over-charge protection voltage (V_{OC}), and the voltage at the VM terminal is above the charge overcurrent protection voltage (V_{CHA}) and below the discharge overcurrent protection voltage (V_{EC}), both the CO and DO terminals of the IC output a high level, causing the MOSFETs for charge control and discharge control to turn on simultaneously. This state is referred to as the "normal operating state." In this state, normal charging and discharging are possible.

Note: When connecting the battery cell for the first time, discharge may not be possible. In this case, short-circuit the VM and VSS terminals, or connect a charger to restore normal operation.

2. Overcharge State

For a battery in normal operation, if during charging the battery voltage across the VDD and VSS terminals exceeds the overcharge protection voltage (V_{OC}) and this condition persists for longer than the overcharge protection delay time (t_{OC}), the IC's CO terminal output voltage changes from high to low, turning off the charging control MOSFET and stopping charging. This state is referred to as the "overcharge state."

The overcharge state can be released under the following two conditions, causing the CO pin output voltage to change from low to high and turning on the MOSFET used for charge control.

(1) When $V_{CHA} < VM < V_{EC}$ and the battery voltage drops below the overcharge release voltage (V_{OCR}), the overcharge condition is released, and normal operation is restored.

(2) If the charger is removed and a load is connected ($VM > V_{EC}$), the overcharge state is released when the battery voltage drops below the overcharge protection voltage (V_{OC}), and normal operation is restored. This function is called the load detection function.

Note: If the charger remains connected after overcharge protection is triggered, the overcharge condition cannot be released even if the battery voltage drops below the overcharge release voltage (V_{OCR}). The overcharge condition is released when the charger is disconnected and the voltage at the VM terminal rises above the charging overcurrent protection voltage (V_{CHA}).

3. Over-discharge Condition

For a battery in normal operation, if during discharge the battery voltage between the VDD and VSS terminals drops below the over-discharge protection voltage (V_{OD}) and this state persists for longer than the over-discharge protection delay time (t_{OD}), the output voltage at the IC's DO terminal changes from high to low, turning off the MOSFET used for discharge control and stopping discharge. This state is referred to as the "over-discharge state."

In the over-discharge state, if the voltage difference between the VDD and -VM terminals drops below 0.1 V (typical value), the current consumption will decrease to the level during sleep mode.

(IPDN), this state is referred to as the "sleep state." When the charger is not connected and the voltage at the VM terminal is ≥ 0.7 V (typical value), the over-discharge state is maintained even if the battery voltage is above V_{OD} . The over-discharge state can be released under the following two conditions:

(1) When a charger is connected, if $VM \leq V_{CHA}$ and the battery voltage is higher than the over-discharge protection voltage (V_{OD}), the over-discharge state is released, and the device returns to normal operation. This function is referred to as the charger detection function.

(2) When a charger is connected or an external trigger is applied, if $V_{CHA} < VM < V_{EC}$, the over-discharge state is released and normal operation is restored when the battery voltage exceeds the over-discharge release voltage (V_{ODR}).

4. Discharge Overcurrent Condition (Discharge Overcurrent Protection and Short-Circuit Protection Functions)

For a battery in normal operation, the IC continuously monitors the discharge current via the VM terminal. If the voltage at the VM terminal exceeds the discharge overcurrent protection voltage (V_{EC}) and this condition persists for longer than the discharge overcurrent protection delay time (t_{EC}), the output voltage at the DO terminal changes from high to low, turning off the MOSFET used for discharge control and stopping the discharge. This state is referred to as the "discharge overcurrent state." Conversely, if the voltage at the VM terminal exceeds the load short-circuit protection voltage (V_{SHORT}) and this condition persists for longer than the load short-circuit protection delay time (t_{SHORT}), the output voltage at the DO terminal also changes from high to low, turning off the MOSFET used for discharge control and stopping the discharge. This state is referred to as the "load short-circuit state."

Conditions for Clearing the Discharge Overcurrent State: "Load Disconnection" and the Discharge Overcurrent State Clear Voltage " V_{DOV} "

In the discharge overcurrent state, the VM pin and the VSS pin inside the chip are connected via the R_{VMS} resistor. However, while a load is connected, the voltage at the VM pin becomes the same as the VDD pin voltage due to the load connection. If the connection to the load is disconnected, the voltage at the VM pin returns to the VSS pin voltage. When the voltage at the VM pin drops below V_{DOV} , the discharge overcurrent state is released.

5. Charging Overcurrent Condition

Protection IC

For a battery in normal operation, if the voltage at the VM terminal falls below the charge overcurrent protection voltage (V_{CHA}) during charging, and this condition persists for longer than the charge overcurrent protection delay time (t_{CHA}), the output voltage at the CO terminal changes from high to low. This turns off the MOSFET used for charge control, stopping the charging process. This condition is referred to as the "charge overcurrent state."

After entering the charging overcurrent protection state, if the charger is disconnected and the voltage at the VM terminal rises above the charging overcurrent detection voltage (V_{CHA}), the charging overcurrent state is released, and the device returns to normal operation.

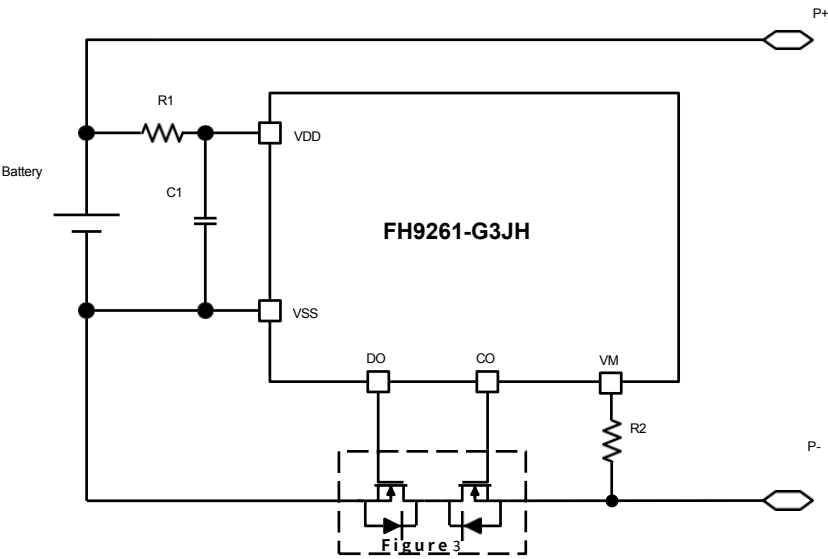
6. Charging a 0V battery function (allowed)

This function is used to recharge a battery that has self-discharged to 0V. When the charger voltage applied between the battery positive terminal (P+) and the battery negative terminal (P-) exceeds the "charger start voltage for charging a 0V battery (V_{OCH})," the gate of the charging control MOSFET is fixed at the potential of the VDD terminal. Since the charger voltage causes the voltage difference between the MOSFET's gate and source to exceed its threshold voltage (V_{th}), the charging control MOSFET turns on, and charging begins. At this point, the discharge control MOSFET remains off, and the charging current flows through its internal parasitic diode. When the battery voltage exceeds the over-discharge protection voltage (V_{OD}), the IC enters normal operating mode.

Note: Please consult the battery manufacturer to determine whether recharging a fully discharged battery is recommended, in order to decide whether charging a 0 V battery is permitted or prohibited.

Application Circuit

Protection IC



Component Designation	Typical Value	Parameter Range	Unit
R1	1	1.0 to 1.5	kΩ
C1	0.1	0.047 ~ 0.220	μF
R2	2	1 to 3	kΩ

Table 5

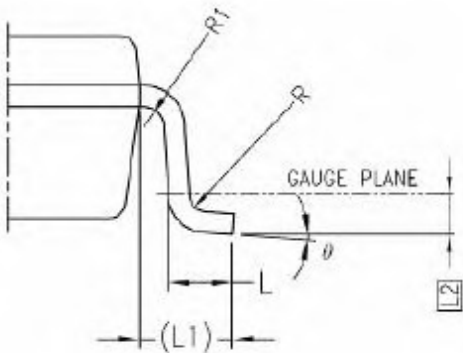
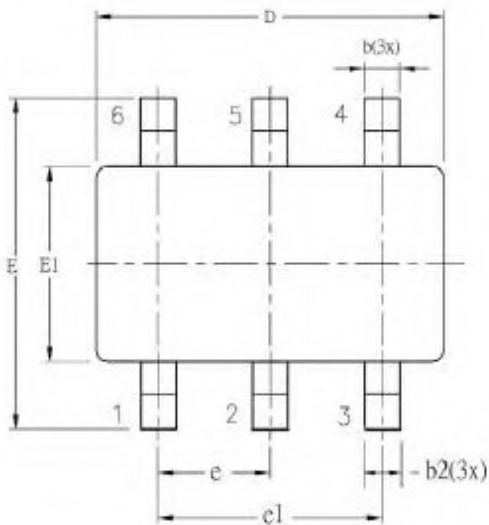
Note:

- 1. The above specifications are subject to change without notice.
- 2. The schematics and parameters of the above ICs are not guaranteed to ensure circuit operation; please set parameters only after thorough testing in actual application circuits.

■ Package Information

Protection IC

SOT23-6 Package: Dimensions in mm.



SYM BOL	ALL DIMENSIONS IN MILLIMETERS		
	MINIMUM	NOMINAL	MAXIMUM
A	-	1.30	1.40
A1	0	-	0.15
A2	0.90	1.20	1.30
b	0.30	-	0.50
b1	0.30	0.40	0.45
b2	0.30	0.40	0.50
c	0.08	-	0.22
c1	0.08	0.13	0.20
D	2.90 BSC		
E	2.80 BSC		
E1	1.60 BSC		
e	0.95 BSC		
e1	1.90 BSC		
L	0.30	0.45	0.60
L1	0.60 REF		
L2	0.25 BSC		
R	0.10	-	-
R1	0.10	-	0.25
θ	0°	4°	8°
θ1	5°	-	15°
θ2	5°	-	15°

